Unit 4: Resampling Techniques

Chapter 8 & 10 in "Statistical Computing with R"

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Overview

1. Introduction
2. Bootstrapping
3. Jackknife
4. Permutation Tests

Introduction

Introduction

- We're going to talk about **the hot buzzword: bootstrapping.**
- Alongside we will talk about **jackknife** and **permutation tests**.

Bootstrapping

Bootstrapping

- Bootstrap methods, introduced by Efron in 1979, are a class of **nonparametric** Monte Carlo methods that estimate the distribution of a population by **resampling**.
- Resampling methods treat an observed sample as a finite population, and random samples are generated (resampled) from it to estimate population characteristics and make inferences about the sampled population.
- It is especially useful when the theoretical distribution of a statistic is unknown or difficult to derive.
- **Bootstrapping is used in machine learning.** For instance, it's used for bagging and random forest (decision trees).
- This class is not about machine learning, so you'll need to take STA314 to learn those.

Bootstrapping

- The term "bootstrap" can refer to nonparametric bootstrap or parametric bootstrap.
- **Parametric bootstrap:** Monte Carlo methods that involve sampling from a fully specified probability distribution. This is what we used in the previous unit.
- **Nonparametric bootstrap:** the distribution is not specified. It generates random samples from the empirical distribution of the sample: F_n. This is what we're doing now.

Bootstrapping

- **In the real world,** we assume samples $X_1, X_2, \ldots, X_n$ with corresponding realisations $x_1, x_2, \ldots, x_n$ occur from an underlying probability distribution F.
- Recall: a statistic is a function of the data, let's denote it as T_n.
- We can calculate a statistic, $T := T_n(x_1, \ldots, x_n)$.

$$F \Rightarrow (x_1, x_2, \ldots, x_n) \Rightarrow T$$

Bootstrapping

- **In the bootstrap world**, we have a finite population composed of our earlier samples $x_1, x_2, \ldots, x_n$ with the probability distribution F_n (the empirical distribution).
- We can extract a bootstrap sample: $x_1^*, x_2^*, \ldots, x_n^*$.
- We can calculate the bootstrap of the statistic: $T^* := T_n(x_1^*, \ldots, x_n^*)$.

$$F_n \Rightarrow (x_1^*, x_2^*, \ldots, x_n^*) \Rightarrow T^*$$

- We can even draw the empirical distribution from the bootstrap sample, F_n^*.

Bootstrapping

In general, we assume:

$$F_n^* \approx F_n \approx F$$

I.e., the ECDF F_n^* of the bootstrap replicates are used to approximate the ECDF of F_n to approximate the cdf F.

Bootstrapping Algorithm

Assume we have the observed sample $x_1, x_2, \ldots, x_n$. The nonparametric bootstrap estimates $\hat{\theta}$ using the following method:

1. For each bootstrap replicate, indexed $b = 1, 2, \ldots, B$:
   - Generate a sample $x^{*(b)} = (x_1^*, \ldots, x_n^*)$ by sampling **with replacement** from $x_1, x_2, \ldots, x_n$.
   - Compute $\hat{\theta}^{(b)}$ from the bth bootstrap sample.
2. The bootstrap estimate $F_{\hat{\theta}}(\cdot)$ is the empirical distribution of the replicates $(\hat{\theta}^{(1)}, \hat{\theta}^{(2)}, \ldots \hat{\theta}^{(B)})$.

Bootstrapping Example 1: Estimate of Standard Error

1. For each bootstrap replicate, indexed $b = 1, 2, \ldots, B$:
   - Generate a sample $x^{*(b)} = (x_1^*, \ldots, x_n^*)$ by sampling **with replacement** from $x_1, x_2, \ldots, x_n$.
   - For each bth bootstrap sample, compute the sample mean $(\overline{x}^*)^{(b)} = \frac{1}{n}\sum_{i=1}^n x_i^*$.
2. The bootstrap estimate of $SE(\overline{x}^*)$ is the sample standard error of the replicates $(\overline{x}^*)^{(1)}, (\overline{x}^*)^{(2)}, \ldots, (\overline{x}^*)^{(B)}$. That is,

$$SE(\overline{x}^*) = \sqrt{\frac{\sum_{b=1}^B [(\overline{x}^*)^{(b)} - \overline{x}']^2}{B-1}}$$

Where $\overline{x}' = \frac{1}{B}\sum_{b=1}^n (\overline{x}^*)^{(b)}$.

Bootstrapping Example 2: Estimate of Bias

1. First, find $\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$, which is the estimate computed from the original observed sample.
2. For each bootstrap replicate, indexed $b = 1, 2, \ldots, B$:
   - Generate a sample $x^{*(b)} = (x_1^*, \ldots, x_n^*)$ by sampling **with replacement** from $x_1, x_2, \ldots, x_n$.
   - For each bth bootstrap sample, compute the sample mean $(\overline{x}^*)^{(b)} = \frac{1}{n}\sum_{i=1}^{n} x_i^*$.
3. The bootstrap estimate of the bias is $Bias(\overline{x}^*) = \frac{1}{B}\sum_{b=1}^{B}(\overline{x}^*)^{(b)} - \overline{x}$.

Recall, bias is defined as:

$$bias(\hat{\theta}) = \mathbb{E}[\hat{\theta} - \theta] = \mathbb{E}[\hat{\theta}] - \theta$$

Bootstrapping Example

Example

Use the "sleepstudy" dataset from the lme4 R package. Do the following:

(a) Try to understand what the dataset is about. Use '??lme4::sleepstudy'

(b) Using nonparametric bootstrapping, estimate the mean of the reaction time.

(c) Compute the bootstrap estimate of the standard error of the sample mean.

(d) Compute the bootstrap estimate of bias of the sample mean.

Solution. Consult the R codes.

Aside: Picking Good R Packages

- Let's be real: most things you learn to compute in undergrad was likely automated by someone else.
- For example, the easystats package makes a lot of data analysis trivial, especially concepts you might see in STA305.
- Similar with bootstrapping; there are external packages that can do just that.
- However, there are too many packages with the bootstrapping tag!
- Try: '??bootstrap' to see what happens!

Question for You!

How can we assess the trustworthiness of an R package?

Aside: Picking Good R Packages

1. **Maintenance and Activity:** check how actively the package is maintained (e.g., regular updates, issue resolution, community activity). Packages that show recent commits and responsiveness to user feedback are generally more reliable.
2. **Package Popularity:** use tools like packageRank to see download counts, indicating user trust and popularity.
3. **Published Papers:** Look for academic publications or citations referencing the package.

Aside: Picking Good R Packages

4. **CRAN and GitHub:** CRAN enforces a review process. It's easier to view code (and live updates) on GitHub. Look for packages that use both.
5. **Functionality Search:** to find packages based on specific functionality or terms, try using library(sos) and findFn("...") to search CRAN documentation for keywords. Keep in mind that search results can vary widely: common terms (e.g., "penalized regression") may yield many results, while more niche terms (e.g., "Firth regression") provide a narrower, more targeted list.
6. **Reverse Dependencies/Suggests:** check if other packages depend on it (reverse dependencies), as packages with several dependencies often reflect broader trust and utility. Common dependencies include Matrix, ggplot2, etc.

Percentile Bootstrapping Confidence Intervals

Bootstrapping is useful when we don't know the sampling distribution. We can attempt to build confidence interval based on percentiles.

Percentile Bootstrap CI

$(\theta^*_{\alpha/2}, \theta^*_{1-\alpha/2})$ where $\theta^*_{\alpha/2}$ and $\theta^*_{1-\alpha/2}$ are sample percentiles based on $\theta^*_1, \ldots, \theta^*_B$.

Percentile Bootstrapping CI Example

Example

Use the sleepstudy dataset from lme4 package to provide a 95% **percentile bootstrap** confidence interval of the mean for reaction time.

Bootstrapping CI

- The percentile bootstrap method is common and appealing since it is easy to compute.
- However it depends on how well the bootstrap sample is actually a good representation of the true distribution. It does not work well if the sample is poor.
- Hence, some people will use the Empirical CI instead.

Empirical Bootstrapping CI

- Bootstrap is useful because resampling gives us an estimate on how the point estimate varies.
- Note: resampling won't improve our point estimation.
- Bootstrap is more robust when approximating the distribution of relative variation.
- Consider the distribution of:

$$\delta = \overline{x} - \mu$$

Empirical Bootstrapping CI

Let $\delta_{1-\alpha/2}$ and $\delta_{\alpha/2}$ represent the critical values of δ. Then,

$$\begin{aligned} 1-\alpha &= \mathbb{P}(\delta_{1-\alpha/2} \leq \delta \leq \delta_{\alpha/2}) \\ &= \mathbb{P}(\delta_{1-\alpha/2} \leq \overline{x} - \mu \leq \delta_{\alpha/2}) \\ &= \mathbb{P}(\overline{x} - \delta_{\alpha/2} \leq \mu \leq \overline{x} - \delta_{1-\alpha/2}) \end{aligned}$$

If we approximate $\delta^* = \overline{x}^* - \overline{x}$ then we have the following CI (replace δ with δ^*:

Empirical Bootstrap CI

$$1-\alpha = \mathbb{P}(\overline{x} - \delta^*_{\alpha/2} \leq \mu \leq \overline{x} - \delta^*_{1-\alpha/2})$$

Empirical Bootstrapping CI - Why This Works

- Note that if we use bootstrapped samples to estimate the mean:

$$\overline{x}^* \approx \overline{x} \approx \mu$$

  we are trying to use the bootstrap to estimate a single parameter value.
- δ^*, δ are both distributions.
- Note that the ecdf converges uniformly to the true cdf.
- We can think of δ^* of being like an ecdf, and δ being the true cdf.
- That's why bootstrapping works well in this case.

Empirical Bootstrapping CI Example

Example

Use the sleepstudy dataset from lme4 package to provide a 95% **empirical bootstrap** confidence interval of the mean for reaction time.

Jackknife

Jackknife

- The jackknife is another resampling method, proposed for estimating bias and standard error.
- The jackknife is like a "leave-one-out" type of cross-validation. (Cross-validation is discussed in STA314).
- It is "simpler" and "faster" than the bootstrap, since we do not draw new (bootstrap) samples, but we leave out one value from the original sample (for each jackknife sample).
- Let $\mathbf{x} = (x_1, \ldots, x_n)$ be an observed random sample, and define the ith jackknife sample $x_{(i)}$ to be the subset of x that leaves out the ith observation x_i.
- That is, $x_{(i)} = (x_1, \ldots, x_{i1}, x_{i+1}, \ldots, x_n)$.

Jackknife

If $\hat{\theta} = T_n(x)$ then define the ith jackknife replicate to be:

$$\hat{\theta}_{(i)} = T_{n-1}(x_{(i)})$$

To get the jackknife estimate, we take the average of the jackknife replicates. That is,

$$\overline{\hat{\theta}_{(.)}} = \frac{1}{n}\sum_{i=1}^{n}\hat{\theta}_{(i)}$$

Jackknife

Example

Compute a jackknife estimate of the sample mean and variance of the reaction time from the `sleepstudy` dataset found in `lme4`. Now, compare this to the sample mean of the reaction times. What do you notice?

Jackknife

Example

Prove that the jackknife mean is equal to the sample mean.

Solution.

$$\overline{x}_{(i)} = \frac{1}{n-1} \sum_{j=1, j \neq i}^{n} x_i = \frac{1}{n-1}(n\overline{x} - x_i)$$

$$\overline{\overline{x}_{(.)}} = \frac{1}{n} \sum_{i=1}^{n} \overline{x}_{(i)} = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{n-1}(n\overline{x} - x_i) = \frac{1}{n} \frac{1}{n-1}(n^2\overline{x} - n\overline{x}) = \overline{x}$$

Jackknife Estimate of Bias

Hence, the Jackknife is not used to improve the theoretical mean, but it can be useful for estimating bias and standard error instead.

Jackknife Estimate of Bias

If $\hat{\theta}$ is a smooth statistic, then $\hat{\theta}_{(i)} = t(F_{n-1}(x_{(i)})$ and the jackknife estimate of bias is:

$$\widehat{bias}_{jack} = (n-1)\mathbb{E}[\hat{\theta}_{(.)} - \hat{\theta}]$$

where $\hat{\theta}$ is the estimate computed from the original observed sample.

Jackknife Estimate of Bias

Example

If $X_1, \ldots, X_n$ arise from a $N(\mu, \sigma^2)$ distribution. Let $\widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$. Compute $\mathbb{E}[\widehat{\sigma^2_{(i)}} - \widehat{\sigma^2}]$ using the following steps:

(a) $\mathbb{E}[\widehat{\sigma^2} - \sigma^2]$

(b) $\mathbb{E}[\widehat{\sigma^2_{(.)}} - \sigma^2]$

(c) Use $\mathbb{E}[\widehat{\sigma^2_{(.)}} - \widehat{\sigma^2}] = \mathbb{E}[\widehat{\sigma^2_{(.)}} - \sigma^2 + \sigma^2 - \widehat{\sigma^2}]$ and part (a), (b) to compute $\mathbb{E}[\widehat{\sigma^2_{(.)}} - \widehat{\sigma^2}]$.

Then, show that:

$$Bias(\hat{\sigma}^2) = (n-1)\mathbb{E}[\widehat{\sigma^2_{(.)}} - \widehat{\sigma^2}].$$

Jackknife Estimate of Bias

$$
\begin{aligned}
\mathbb{E}[\widehat{\sigma^2} - \sigma^2] &= \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}(x_i - \mu + \mu - \overline{x})^2\right] - \sigma^2 \\
&= \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}(x_i - \mu)^2 + \sum_{i=1}^{n}2(x_i - \mu)(\mu - \overline{x}) + \sum_{i=1}^{n}(\mu - \overline{x})^2\right] - \sigma^2 \\
&= \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}(x_i - \mu)^2 - 2(\mu - \overline{x})\sum_{i=1}^{n}(x_i - \mu) + n(\mu - \overline{x})^2\right] - \sigma^2 \\
&= \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}(x_i - \mu)^2 - 2n(\mu - \overline{x})^2 + n(\mu - \overline{x})^2\right] - \sigma^2
\end{aligned}
$$

Jackknife Estimate of Bias

$$
\begin{aligned}
\mathbb{E}[\widehat{\sigma^2} - \sigma^2] &= \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}(x_i - \mu)^2 - n(\mu - \overline{x})^2\right] - \sigma^2 \\
&= \sigma^2 - \mathbb{E}\left[(\mu - \overline{x})^2\right] - \sigma^2 \\
&= -\mathbb{E}\left[\mu^2 - 2\mu\overline{x} + \overline{x}^2\right] \\
&= -(\mu^2 - 2\mu^2 + \mathbb{E}[\overline{x}^2]) \\
&= -(-\mu^2 + (\mathbb{E}[\overline{x}]^2 + \mathbb{V}ar[\overline{X}])) \\
&= -(-\mu^2 + (\mu^2 + \frac{\sigma^2}{n})) \\
&= -\frac{\sigma^2}{n}
\end{aligned}
$$

Jackknife Estimate of Bias

Note that the estimate of $\widehat{\sigma^2_{(i)}}$ is similar to $\widehat{\sigma^2}$ except we leave out the realisation x_i. I.e., we instead are averaging out of $n-1$ samples instead of n. Thus,

$$\mathbb{E}[\widehat{\sigma^2_{(.)}} - \sigma^2] = -\frac{\sigma^2}{n-1}$$

And therefore,

$$\begin{aligned}
\mathbb{E}[\widehat{\sigma^2_{(.)}} - \widehat{\sigma^2}] &= \mathbb{E}[\widehat{\sigma^2_{(.)}} - \sigma^2 + \sigma^2 - \widehat{\sigma^2}] \\
&= \mathbb{E}[\widehat{\sigma^2_{(.)}} - \sigma^2] - \mathbb{E}[\widehat{\sigma^2} - \sigma^2] \\
&= -\frac{\sigma^2}{n-1} - \left(-\frac{\sigma^2}{n}\right) \\
&= \frac{-n\sigma^2 + (n-1)\sigma^2}{n(n-1)} \\
&= \frac{-\sigma^2}{n(n-1)}
\end{aligned}$$

Jackknife Estimate of Bias

$$\mathbb{E}[\widehat{\sigma^2_{(\cdot)}} - \widehat{\sigma^2}] = \frac{\mathbb{E}[\widehat{\sigma^2} - \sigma^2]}{n-1}$$
$$= \frac{Bias(\hat{\sigma}^2)}{n-1}$$

Rearranging for $Bias(\hat{\sigma}^2)$, we have that:

$$Bias(\hat{\sigma}^2) = (n-1)\mathbb{E}[\widehat{\sigma^2_{(\cdot)}} - \widehat{\sigma^2}].$$

Jackknife

Example

Consider the reaction time data from the `sleepstudy` dataset in the `lme4` package. Compute the jackknife estimate of the bias of the following estimator for the variance of the reaction times:

$$\widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

Jackknife Estimate of Standard Error

Jackknife Estimate of Standard Error

If $\hat{\theta}$ is a smooth statistic, then a jackknife estimate of standard error is:

$$\widehat{SE}_{jack} = \sqrt{\frac{n-1}{n}\sum_{i=1}^{n}\left(\hat{\theta}_{(i)} - \overline{\hat{\theta}_{(.)}}\right)^2}$$

Remark: when it comes to the origin, the factor $\frac{n-1}{n}$ is arbitrary, and thus we will omit the "proof".

Issue with Jackknife

- The jackknife assumes that the statistic being estimated is a smooth function of the data.
- If the statistic is not smooth, then removing one observation can significantly change the result.
- For example, if you try to compute the jackknife estimate of the standard error of the median, then it would fail since the median statistic is not a smooth function.

Median

Let $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ represent a random sample from any distribution, and let their corresponding order statistics be denoted as $X_{1:n} \leq X_{2:n} \leq \cdots \leq X_{n:n}$. Then,

$$\text{median}(\mathbf{X}) = \begin{cases} \frac{X_{(n/2+1):n}}{2} & n \text{ is odd,} \\ \frac{X_{(n/2):n} + X_{(n/2+1):n}}{2} & n \text{ is even.} \end{cases}$$

Issue with Jackknife

Example

Compute the jackknife estimate of the **standard error** of the **median** from a random sample of 10 integers from 1, 2, . . . , 100 and see how it **fails.**

Solution.

Permutation Tests

Permutation Tests

- Permutation tests are based on resampling **without replacement**.
- Permutation tests are often applied as a nonparametric hypothesis test to compare two distributions.
- In a permutation test, the test statistic is calculated for all possible samples of $x's$ and $y's$ drawn without replacement from the combined data.

Permutation Test for Equal Distributions

- Suppose that $\mathbf{X} = (X_1, \ldots, X_n)$ and $\mathbf{Y} = (Y_1, \ldots, Y_m)$ are independent random samples from distributions F and G, respectively, and we wish to test the hypothesis $H_0 : F = G$ vs. the alternative $H_1 : F \neq G$.
- Under the **null** hypothesis, the pooled sample $X \cup Y$ all come under the same distribution F (or G).
- The most widely applied permutation test of equal distributions for univariate distributions is the **Kolmogorov-Smirnov (K-S) statistic**.

Kolmogorov-Smirnov (K-S) Statistic

Let F_n be the ecdf of the first sample and G_m be the ecdf of the second sample. The Kolmogorov-Smirnov (K-S) statistic D is the maximum absolute difference between the ecdf's of the two samples:

$$D = \sup_{1 \leq i \leq N} |F_n(z_i) - G_m(z_i)|$$

Permutation Test for Equal Distributions

Example

Generate $n = 30$ samples from an *Exponential*$(\theta = 1/3)$ and a *Gamma*$(\alpha = 2, \beta = 1/5)$ distribution.

(a) Directly compute the Kolmogorov-Smirnov (K-S) statistic D, and compare the result to `ks.test()`. Call this `D0`.

(b) Pool the samples together. Then, for $R = 999$ replicates, randomly assign the pooled samples into two disjoint groups without replacement. Re-compute a new Kolmogorov-Smirnov (K-S) statistic D_i for $i = 1, 2, \ldots, R$.

(c) Intuitively, if we have that $D_i \geq D_0$, this means the **permuted samples produce an equally or more extreme separation than the original**, meaning the original D_0 is not surprisingly large, thus failing to reject the null hypothesis. Compute a p-value by averaging over the number of cases where $D_i \geq D$.